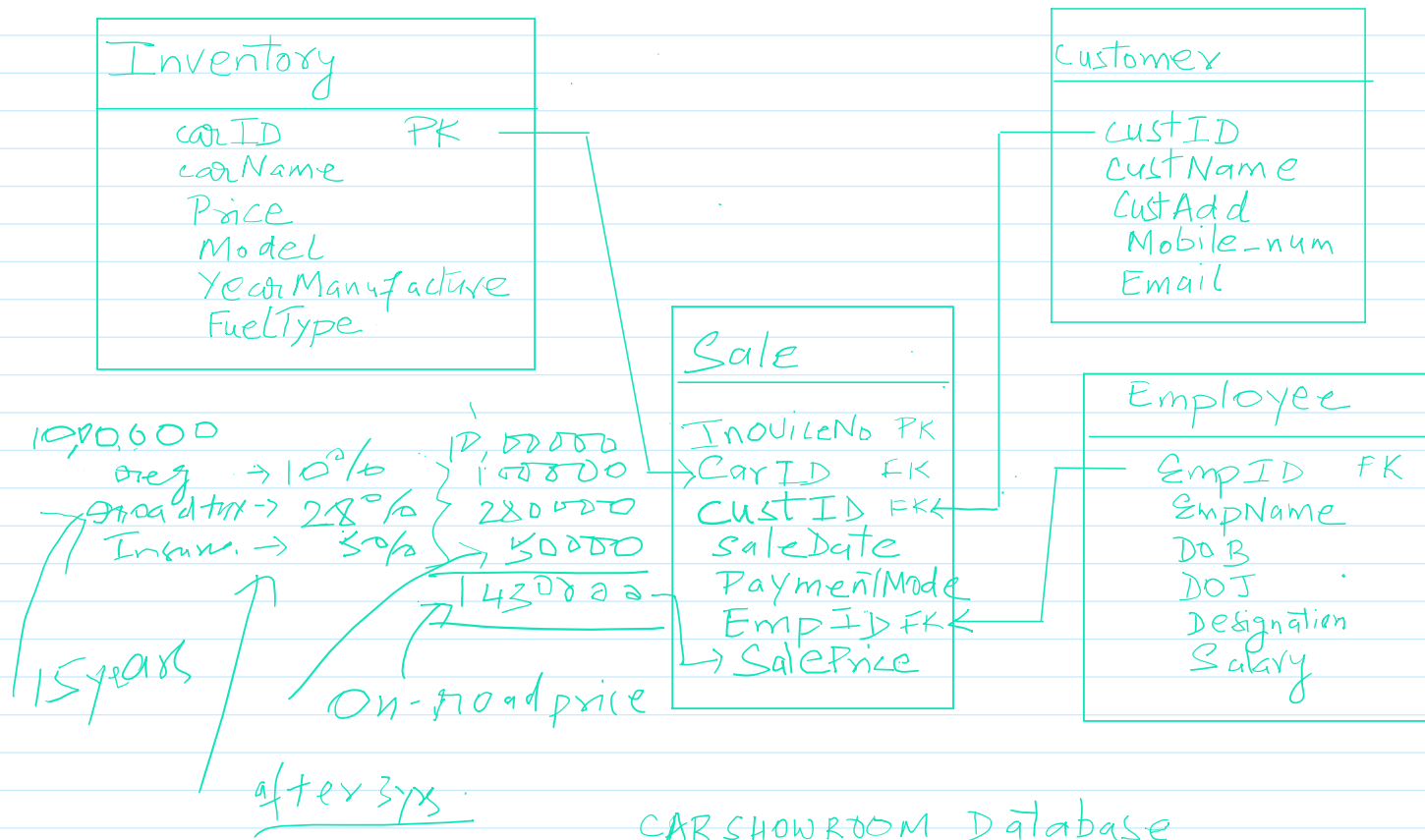


Functions in SQL

08 September 2026 09:24

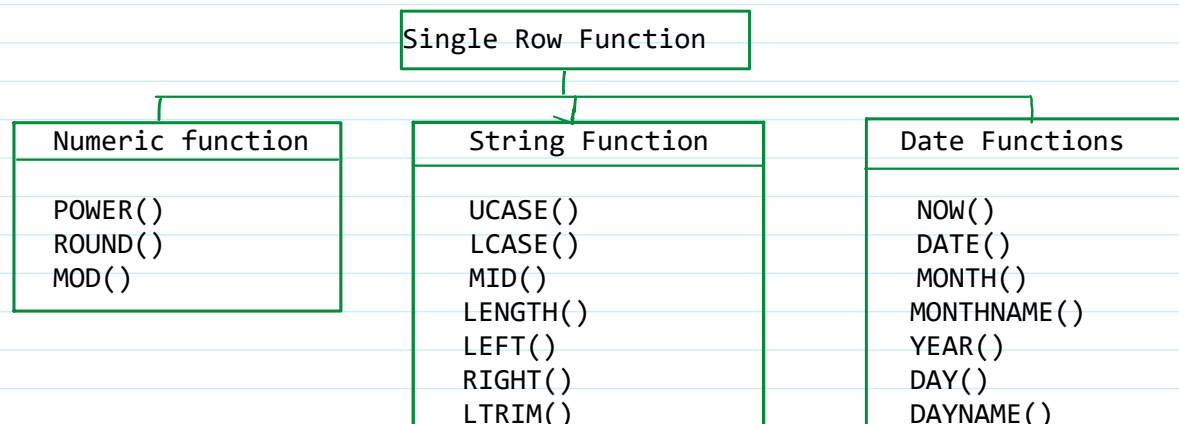
We will understand how to use single row function, multiple row function, group records based on some criteria, and working on multiple tables using SQL.



- * Create database.
- * Create relations (tables)
- * insert values in the relations (tables).

Single Row Functions

These are also known as Scaler Functions. Single row functions are applied on a single value and return a single value. Different single row functions under three categories - numeric (math) String, Date and time.



LEFT()
RIGHT()
LTRIM()
RTRIM()
TRIM()

YEAR()
DAY()
DAYNAME()

Math function except numeric value as input and return a numeric value as a result. String function accept character value as the input and return either character or numeric values as output. Date and time function except date and time value as input and return numeric or string or date and time as output.

Math Functions

Function	Description	Example with output
POWER(X, Y)	Calculates X to the power Y.	<pre>mysql> SELECT POWER(2, 3); +-----+ POWER(2, 3) +-----+ 8 +-----+ 1 row in set (0.144 sec) mysql> SELECT POWER(4, 0.5); +-----+ POWER(4, 0.5) +-----+ 2 +-----+ 1 row in set (0.021 sec) mysql> SELECT POWER(9, 0.5); +-----+ POWER(9, 0.5) +-----+ 3 +-----+ 1 row in set (0.032 sec) mysql> SELECT POWER(8, 1/3); +-----+ POWER(8, 1/3) +-----+ 1.999861375368307 +-----+ 1 row in set (0.022 sec) mysql> SELECT POWER(8, 0.33); +-----+ POWER(8, 0.33) +-----+ 1.9861849908740719 +-----+</pre>

```

1 row in set (0.137 sec)

mysql> SELECT POWER(8, 0.3);
+-----+
| POWER(8, 0.3) |
+-----+
| 1.8660659830736148 |
+-----+
1 row in set (0.005 sec)

mysql> SELECT POWER(8,
0.34);
+-----+
| POWER(8, 0.34) |
+-----+
| 2.027918959580058 |
+-----+
1 row in set (0.007 sec)

mysql> SELECT SQRT(16)
-> ;
+-----+
| SQRT(16) |
+-----+
| 4 |
+-----+
1 row in set (0.029 sec)

mysql> SELECT ROUND(283.2);
+-----+
| ROUND(283.2) |
+-----+
| 283 |
+-----+
1 row in set (0.142 sec)

mysql> SELECT ROUND(283.5);
+-----+
| ROUND(283.5) |
+-----+
| 284 |
+-----+
1 row in set (0.009 sec)

mysql> SELECT
ROUND(283.5671, 2);
+-----+
| ROUND(283.5671, 2) |
+-----+
| 283.57 |
+-----+
1 row in set (0.005 sec)

mysql> SELECT MOD(25, 7);
+-----+

```

ROUND(N, D) Rounds of number N to D number of decimal places.
Note: If D equal to zero then it round off the number to the nearest integer.

MOD(A, B) Return the remainder after dividing number A by number B.

MOD(25, 7)
4
1 row in set (0.023 sec)

Homework:

1. Calculate GST as 18% of price and display the result after rounding it off to one decimal place.
2. Add a new column FinalPrice to the table inventory which will have the values as sum of price and 18% of GST.
3. Calculate and display the amount to be paid each month (in multiples of 1000) which is to be calculated after dividing the final price of the car into ten installments.
4. After dividing the amount into EMI's find out the remaining amount to be paid immediately by performing modular division.
5. Let us now add a new column Commission to the sales table. The column Commission should have total length of 7 in which two decimal places to be there.
6. Let us now calculate Commission for sales agents as 12% of the sale price. Insert the values to the newly added column Commission and then display records of the sales where Commission is greater than 73,000.
7. Display invoice number, sale price and commissions such that Commission value is rounded off to 0.